

6. (3.5 points)

Given the following information:

Age-to-Age Development Factors

Accident Year	12-24 Months	24-36 Months	36-48 Months	48-60 Months
2011	1.600	1.375	1.091	1.125
2012	5.000	1.100	2.000	
2013	2.833	1.588		
2014	2.091			

- Assume that $T = r [(n - 2) / (1 - r^2)]^{1/2}$
- The following table displays the t -statistic for 0.9 at various degrees of freedom:

Degrees of Freedom	<u>1</u>	<u>2</u>	<u>3</u>
t -statistic	6.314	2.920	2.354

a. (2 points)

Using Venter's correlation test, determine whether the correlation between the pair of development factor columns (12-24 Months) and (24-36 Months) is significant at the 10% level.

b. (1 point)

Calculate Spearman's rank correlation coefficient for the triangle as a whole.

c. (0.5 point)

Briefly describe two reasons why it may be more appropriate to consider the correlation of a loss development triangle as a whole instead of correlations between pairs of columns.

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QUESTION 6

TOTAL POINT VALUE: 3.5

LEARNING OBJECTIVE: A2: Estimate parameters and unpaid claims using claims development models related to loss reserving methods such as chain ladder, Cape Cod, chain ladder plus calendar-year effects, and Bornhuetter-Ferguson.

SAMPLE ANSWERS

Part a: 2 points

Sample Answer 1

$$r = \frac{\sum (X - E[X]) \times (Y - E[Y])}{(\sum (X - E[X])^2 \times \sum (Y - E[Y])^2)^{.5}}$$

12-to-24	24-to-36
Months	Months

AY	X	Y	(X - E[X])	(Y - E[Y])	(X - E[X]) ²	(Y - E[Y]) ²	(X - E[X]) × (Y - E[Y])
2011	0.6000	0.3750	(1.5443)	0.0207	2.3850	0.0004	(0.0319)
2012	4.0000	0.1000	1.8557	(0.2543)	3.4435	0.0647	(0.4720)
2013	1.8330	0.5880	(0.3113)	0.2337	0.0969	0.0546	(0.0727)
Mean	2.1443	0.3543		Σ	5.9254	0.1197	(0.5766)

$$r = \frac{-0.5766}{(5.9254 \times 0.1197)^{.5}}$$

$$r = -0.6846$$

n= 3

$$DF = n - 2$$

DF = 1

$$t\text{-statistic} = 6.3140$$

$$T = r \times [(n - 2) / (1 - r^2)]^{.5}$$

$$T = -0.6846 \times [(3 - 2) / (1 - (-0.6846)^2)]^{.5}$$

T= -0.9393

$$|T| = 0.9393$$

Since $0.9393 < 6.3140$, the correlation between the columns is not significant.

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Sample Answer 2

$$r = \frac{E[XY] - E[X] \times E[Y]}{\sigma_X \times \sigma_Y}$$

	12-to-24 Months	24-to-36 Months		(X - E[X]) ²	(Y - E[Y]) ²
AY	X	Y	XY		
2011	0.6000	0.3750	0.2250	2.3850	0.0004
2012	4.0000	0.1000	0.4000	3.4435	0.0647
2013	1.8330	0.5880	1.0778	0.0969	0.0546
Mean	2.1443	0.3543	0.5676	Σ 5.9254	0.1197
				n 3	3
				Variance	1.9751
				Standard Dev.	1.4054

$$r = \frac{0.5676 - 2.1443 \times .3543}{(1.4054 \times 0.1998)}$$

$$r = -0.6846$$

$$n = 3$$

$$DF = n - 2$$

$$DF = 1$$

$$\text{t-statistic} = 6.3140$$

$$T = r \times [(n - 2) / (1 - r^2)]^{.5}$$

$$T = -0.6846 \times [(3 - 2) / (1 - (-0.6846)^2)]^{.5}$$

$$T = -0.9393$$

$$ITI = 0.9393$$

Since 0.9393 < 6.3140, the correlation between the columns is not significant.

Sample Answer 3

$$r = \frac{E[XY] - E[X] \times E[Y]}{((E[X^2] - E[X]^2) \times (E[Y^2] - E[Y]^2))^{.5}}$$

	12-to-24 Months	24-to-36 Months			
AY	X	Y	XY	X ²	Y ²
2011	0.6000	0.3750	0.2250	0.3600	0.1406

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2012	4.0000	0.1000	0.4000	16.0000	0.0100
2013	1.8330	0.5880	1.0778	3.3599	0.3457
Mean	2.1443	0.3543	0.5676	6.5733	0.1655
Mean ²	4.5982	0.1256			
$r = \frac{0.5676 - 2.1443 \times .3543}{((6.5733 - 4.5982) \times (0.1655 - 0.1256))^{.5}}$					
$r = -0.6846$					
$n = 3$					
$DF = n - 2$					
$DF = 1$					
$t\text{-statistic} = 6.3140$					
$T = r \times [(n - 2) / (1 - r^2)]^{.5}$					
$T = -0.6846 \times [(3 - 2) / (1 - (-0.6846)^2)]^{.5}$					
$T = -0.9393$					
Since 0.9393 < 6.3140, the correlation between the columns is not significant.					
Part b: 1 point					
$T_k = 1 - \frac{S_k}{n(n^2 - 1)/6}$					
	12-to-24 Months	24-to-36 Months	Rank X	Rank Y	Rank (X-Y) ²
AY	X	Y	X	Y	
2011	1.600	1.375	3	2	1
2012	5.000	1.100	1	3	4
2013	2.833	1.588	2	1	1
			S ₂ =	Σ	6
$l = 5$					
$n = 3$					
$k = 2$					
$l - k - 1 = 2$					
$T_2 = 1 - \frac{6}{3 \times (3^2 - 1)/6}$					

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$T_2 =$	-0.50				
	24-to-36 Months	36-to-48 Months	Rank X	Rank Y	Rank (X-Y) ²
AY	X	Y	X	Y	
2011	1.375	1.091	1	2	1
2012	1.100	2.000	2	1	1
			$S_3 =$	Σ	2
$l =$	5				
$n =$	2				
$k =$	3				
$l - k - 1 =$	1				
$T_3 =$	$1 - \frac{2}{2 \times (2^2 - 1)/6}$				
$T_3 =$	-1.00				
$T =$	$\frac{\Sigma(l-k-1) \times T_k}{\Sigma(l-k-1)}$				
$T =$	$\frac{-0.5 \times 2 + -1 \times 1}{(2+1)}$				
$T =$	$\frac{-2}{3}$				
$T =$	-0.67				
Part c: 0.5 point					
<i>Sample Answers</i>					
• Avoid an accumulation of error probabilities • More important to know whether correlations globally prevail than to find a small part of the triangle with correlations. • At a 10% level of significance, 10% of the pairs of columns could show up as significant just by random happenstance. A single significant correlation would not be a strong indication of correlation within the triangle.					
EXAMINER'S REPORT					
Overall, many candidates performed very well on this question. Based on the knowledge statements, candidates should know key assumptions of the chain ladder models and how to test these assumptions. The question tests the candidates' understanding of approaches to test whether age-to-age factors are independent.					

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Part a

Candidates were expected to execute the mechanics of Venter's correlation test, calculating all components of the formula. Candidates performed well on this part.

The most common mistake was not considering this a two tailed test and using $-0.94 < 6.314$ instead of $|-0.94|$ or $-6.314 < -0.94 < 6.314$.

Part b

Candidates were expected to execute the mechanics of calculating the Spearman rank correlation coefficient, calculating all components of the coefficient. Candidates performed well on this advanced correlation test. The most common mistakes included calculation errors or missing the T_k formula.

Part c

Candidates were expected to know how to compare correlations for the data triangle as whole to column-by-column correlations. This was a somewhat challenging question as most candidates would either present a single reason or provide two reasons that were paraphrases of each other. An example of this would be stating that comparing columns would lower the credibility of the measured correlations, and also stating that observed correlations in adjacent columns could be due to random variation.